9.3 Collecting Data

Algebra II 9.3 · Student sheet

Name _____ Period ____ Date _____

I can…

- identify types of samples and tell whether a sample is biased.
- identify the method of data collection.
- recognize biased survey questions and rewrite them.

Vocabulary · fill in the missing word

Random sample — every member of the _____ has an equal chance of being chosen

Self-selected sample — members _____

Systematic sample — a rule is used, such as every 10th _____

Stratified sample — the _____ is divided into groups, and members are randomly chosen from each group

Cluster sample — the _____ is divided into groups, and all members of some randomly chosen groups are surveyed

Convenience sample — only _____ who are easy to reach are chosen

Bias — a sample or question that overrepresents or _____ part of the population

Methods — experiment, _____ study, survey, simulation

Part 1 · Which sample?

A student council wants to know what all students think about a new schedule. Name the type of sample and decide whether it is biased.

Plan	Type of sample	Biased?
1. Every 20th name on the roster		
2. Students who reply to an email		
3. 10 random students from each grade		
4. Everyone in 4 randomly chosen classes		
5. Friends of the student council		
6. 100 names drawn from all students		

Rewrite one biased plan to make it better:

Part 2 · Notes: fill in the blanks

Stratified: random members from _____ group. **Cluster:** _____ members of some groups.

Experiment: apply a _____. Observational study: _____ without changing anything.

A biased question _____ people toward an answer.

Part 3 · Examples (work them with the class)

Example 1

Identify the type of sample.

a. every 10th student who enters the gym

b. all students in 3 randomly chosen homerooms

c. 20 randomly chosen students from each grade

d. students who answer a poll on the school website

Example 2

A town wants to know whether residents support building a new park. It surveys people at an existing park. Is the sample biased? Explain.

Example 3

Identify the method of data collection.

a. A researcher records how many cars run a red light at an intersection.

b. A computer flips 1,000 virtual coins.

c. Doctors give half of the patients a new drug.

d. A questionnaire is mailed to customers.

Example 4

Is the question biased? If so, rewrite it.

“Don’t you agree that the school day is too long?”

Part 4 · Partner practice

Partner 1

Identify the type of sample.

a. A coach randomly picks 2 players from each grade.

b. A teacher asks the first 10 students who arrive.

Partner 2

Is the question biased? If so, rewrite it.

“Wouldn’t a new gym be better for our students than a new library?”